

Evaluating Diagnostic Edge Devices for Remote Clinics

Introduction

There is a compromise to be made between performance, power usage and hardware durability in the face of unpredictable power and network outages when using portable edge diagnostic devices in rural off-grid clinics. At the same time these edge systems have to deal with the ingestion of biomedical sensor data in real time, machine learning inference based on the data to detect abnormalities, secure local logging, and batch transfer to cloud systems with minimal network connections. Two architectures Device Alpha, that is ARM Cortex-A55 and Device Beta, that is RISC-V are evaluated based on memory hierarchy, task scheduling capacity and power resilience which helps us to detect the best design for a resource-constrained healthcare setting.

Background / Related Work

The main design compromise in embedded edge AI is that of memory bandwidth/caching and energy consumption. As per memory hierarchy principle, smaller on chip memory structures, with SRAM (L1/L2 caches) help to reduce the average latency to access the memory, while off chip memory accesses, with DRAM, are more likely to affect system latency and energy consumption (Patterson & Hennessy, 2017). The CPUs inside the Device Alpha are quad-core ARM Cortex-A55 that have 32 KB of instruction and data cache, and a 1 MB unified L2 cache (~80 ns latency, up to 25.6 GB/s bandwidth), (ARM Limited, 2021). Device Beta has a dual-core RISC-V implementation of the SiFive U740R (125K 8-stage) with a larger 2 MB shared L2 cache (~12 cycles latency), and uses 512 MB LPDDR3 memory (~17 GB/s and 90-100 ns latency) (SiFive Inc., 2023). Edge reliability is also determined by the storage technology. Consumer grade NAND flash (as used in Alpha's 128 GB SSD) has high density, and low endurance. (~3,000 P/E cycles; ~10-year retention) (Cypress Semiconductor, 2019). In contrast, NOR flash (which is used in Beta's 64 MB storage) is very durable (~100,000 P/E cycles; ~20-year retention), and is well suited for critical logging applications — albeit with capacity limitations (Cypress Semiconductor, 2019).

Methodology / System Design

This analysis evaluates both platforms using architectural specifications, workload memory footprints, and power profiles under typical rural clinic workloads.

Assumption	Quantitative Value	Evidence / Source
Diagnostic ML Model Size	50–100 MB	MobileNetV3 Edge Healthcare Models
Inference Latency	~150 ms	Embedded Inference Benchmarks
Sensor Data per Patient	10 MB / session	Standard Clinical Screening Protocol
Daily Logging / Upload Volume	300 MB / day	30 Patients/Day Deployment Estimate
Autonomous Offline Window	7–14 days	Remote Rural Clinic Operational Requirement

To overcome the performance and memory issues several architectural optimizations are proposed:

- Batch Inference Scheduling: In order to relieve the bandwidth pressure and dynamic energy consumption, batch scheduling to off-chip DRAM, which places 4–8 patient diagnostic samples in a batch where overhead of loading the weights can be amortized in the cache, can be performed.
- Allocating a circular buffer in the shared L2 cache groups diagnostic logs together to flush to persistent storage in 1- KB chunks, which reduces the frequency of writing diagnostic logs to persistent flash memory (Cypress Semiconductor, 2019).
- The reduced working memory footprint by ~75% and the lowered intermediate activation bandwidth have resulted in lower energy consumption with material improvement in model quantization to 8-bit integers (INT8) [cite: 1, 4].
- Using 4-core parallelism and by setting the batch size to 4, Device Alpha is able to execute ~6.7 inferences/s at 5.0 W (~0.75 J/inference) with batch inference. When running Device Beta with 2 cores at 1.8 W (~0.54 J / inference) it runs sequentially at approximately 3.3 inferences per second, which makes it possible to deploy directly on a compact solar cell battery box.

Discussion

The architecture trade-offs are due to the differing needs. Devices Alpha offers better concurrency and storage capacity (128 GB stores ~426 days worth of logs), but with a 5.0 W power requirement, it requires more solar. Also, NAND flash memory on Device Alpha could wear out after 3-5 years when logging continuously (Cypress Semiconductor, 2019). It's got terrific power durability (1.8 W) and write endurance (~100k P/E cycles), but device Beta has only 512 MB of DRAM and 64 MB of NOR flash chips, which are relatively trivial when considering their ability to multitask or local retention (~7 days uncompressed) (SiFive Inc., 2023).

Device Beta, on sustainability terms, runs on 15.8 KWH throughout the year while device Alpha runs on 43.8 KWH throughout the year (reduction of 64%), thus reducing e-waste and extending the life of the battery (IEEE, 2025a). Moreover, RISC-V's open ISA facilitates modular repairability and sourcing of hardware locally, avoiding the RFP/LOI process that introduces hardware obsolescence, and helps to enhance healthcare equity in low resource areas (Patterson & Hennessy, 2017). An optimal design with hybrid architecture wherein Alpha quad-core processor and LPDDR4 Memory is paired with industrial grade NOR/eMMC Storage and cache resident logging.

Conclusion

Due to its low power envelope (1.8 W), and proven long-term viability of its NOR flash, Device Beta is well-suited for off-grid clinics, with the capacity to reduce memory requirements with batching and INT8 quantization (IEEE, 2025b). Device Alpha is a device with a higher amount of computational throughput, but is also susceptible to flash wear and power pullout. In the future implementation, there is a need to look into dynamic model selection, and hardware-software co-design respectively to optimize the diagnostic accuracy and optimize for a sustainable environment (IEEE, 2025b).

Reference

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